# Ali Akbar

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## PROFESSIONAL SUMMARY

Hands-on experience building RAG pipelines, LLM-powered agents with observability, and multimodal systems using LangChain, LangGraph, LangSmith, PyTorch, and vector databases. Research background spanning Machine Learning and Computer Vision across 4 published and ongoing research works. Runner-up (top 2) at the ACM UbiComp 2021 International Nurse Care Activity Recognition Challenge. Acknowledged contributor on an ACL Findings 2025 paper.

## EDUCATION

**University of Dhaka (Website)** **Jan. 2019 - April 2024**
*Bachelor of Science (B.Sc.) in Electrical and Electronic Engineering* ***CGPA: 3.52(out of 4.00)***

## TECHNICAL SKILLS

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| **Languages** | C, Java, Python, MATLAB, MySQL, LaTeX. |
| **ML/DL Frameworks** | PyTorch, TensorFlow, Scikit-learn, OpenCV, MLflow, DVC, FastAPI (learning). |
| **LLM & Agentic AI** | LangChain, LangGraph, LangSmith, Pydantic, FAISS, Pinecone, HuggingFace, LLM APIs, RAG, Advanced RAG, Prompt Engineering. |
| **Data Science Tools** | Pandas, NumPy, Matplotlib, MS Excel. |
| **Other Tools** | Git, Docker, Streamlit. |

## EXPERIENCES

**Research Assistant | Data & Design Lab**, dept. of CSE, University of Dhaka **Feb 2025 – Dec 2025**

- Collected real-time field data from substations, developed data analytics solutions, preprocessed and transformed electricity-sector datasets into structured formats for system integration and data-driven decision-making, and contributed to the development of *Biddyut Bandhu*, a domain-specific chatbot.

**Volunteer ML Researcher | icddr,b** **May 2026 – Present**

- Developing a Linux-based shotgun metagenomics pipeline using SRA Toolkit, KneadData, and MetaPhlAn to acquire, preprocess, and generate taxonomic abundance profiles from publicly available healthy-control gut microbiome sequencing data. Integrating these data with organization's existing cohort of 67 IBD patient samples to support comparative microbiome analysis and downstream machine learning studies for identifying disease-associated microbial signatures.

## PROJECTS

**OCR-Based Handwritten Scanned Document Extraction, Summarization, and Retrieval System.**

- Built an end-to-end pipeline for processing handwritten scanned documents — extracting text via OCR from images and PDFs, chunking content semantically, and indexing into a FAISS vector store for retrieval.
- Implemented a RAG-based QA system returning contextual answers with source references (page numbers and relevant text chunks), enabling document-grounded question answering over handwritten documents.
- Designed a Human-in-the-Loop (HITL) feedback mechanism allowing users to iteratively review, correct, and finalize auto-generated CASE FACT summaries until satisfaction — reducing manual summarization effort.

**Conversational AI Chatbot with Memory Using LangGraph and Streamlit.**

- Built a multi-turn conversational AI chatbot using LangGraph's StateGraph and MemorySaver for graph-based state management and tool calling, enabling context-aware responses across extended conversations.
- Implemented real-time token streaming via Groq inference backend, delivering responses progressively to the Streamlit frontend for a responsive, production-like user experience.
- Architected conversation state as a persistent graph workflow with sidebar controls for memory reset — a more scalable pattern than chain-based memory for agentic applications.

**Retrieval-Augmented Generation (RAG) System for Multi-Source Website Question Answering.**

- Built a RAG pipeline that ingests and fuses content from up to 3 simultaneous user-provided URLs—distinct from document-based RAG in handling live, heterogeneous web sources with varying content lengths.
- Processed and embedded web content using HuggingFace embedding models into a FAISS offline vector store; performed semantic similarity search to retrieve context chunks and generate source-referenced answers.
- Tracked and measured end-to-end pipeline latency using LangSmith; recorded average response time of nearly 28 seconds, with variance attributed to web content length and retrieval complexity.

## APPLIED ML PROJECTS & RESEARCH

**Nurse Care Activity Recognition from Accelerometer Sensor Data Using Fourier- and Wavelet-based Features.** *(May, 2021)*

- Engineered time-frequency features from raw accelerometer signals using FFT and DWT, preceded by a full preprocessing pipeline — data cleaning, resampling, windowing, and bandpass filtering.
- Applied Variance Threshold, ANOVA, and Chi-square feature selection to reduce dimensionality before training and comparing multiple ML classifiers across 25 nurse activity classes.
- Achieved 87% classification accuracy using LightGBM, outperforming other evaluated classifiers; work contributed to a runner-up finish at the ACM UbiComp 2021 Nurse Care Activity Recognition Challenge.

**End-to-End micro-bacterial segmentation and detection using ViT-based hybrid model.**

- Proposed a Max-ViT backbone-based Mask R-CNN for end-to-end microbacterial instance segmentation on the open-source M-ROSE dataset, replacing the standard CNN backbone with a Vision Transformer architecture.
- Outperformed the evaluated baselines — ResNet50 Mask R-CNN, Mask2Former, and YOLOv11 — in case of mAP 79.31% and F1-score 70.01% under 5-fold cross-validation.
- Applied ROI Activation CAM (Region-of-Interest Class Activation Map) for model interpretability, to localize spatially the intra-object regions most influential to segmentation predictions.

**Integrating Gut Microbiome Taxonomic Profiles with Supervised Machine Learning for IBS (Irritable Bowel Syndrome) Diagnosis.**

- Curated a hybrid dataset of 1,109 gut microbiome samples (1,093 from NCBI + 16 newly sequenced Bangladeshi samples) spanning IBS patients and healthy controls — introducing local population representation absent from prior public datasets.
- Processed raw DNA sequences through the CZID bioinformatics pipeline for human DNA removal and quality control, followed by taxonomic feature extraction and multi-step preprocessing for ML readiness.
- Optimized hyperparameters using `Optuna` (Bayesian optimization) under 5-fold cross-validation; trained and compared multiple classifiers evaluated on Accuracy, F1, MCC, and AUROC.
- Logistic Regression outperformed all evaluated models with 79.73% accuracy and MCC 0.5876, indicating moderate-to-strong discriminative power beyond class imbalance; manuscript in preparation for journal submission.

**Thinking Like a Botanist: Challenging Multimodal Language Models with Intent Driven Chain-of-Inquiry**

- Contributed large-scale data collection for PlantInquiryVQA, a multi-step expert-style VQA framework using a Chain of Inquiry (CoI) approach for sequential plant disease diagnosis—work acknowledged in the published paper.
- Dataset comprised 24,964 plant images and 138,078 QA pairs with annotated reasoning chains and visual grounding, supporting evaluation of SOTA MLLMs including Qwen, Grok, Mistral, Gemma etc.
- Models evaluated on both linguistic metrics (F1, BLEU-4, ROUGE-L) and domain-specific metrics (Clinical Factuality, Disease Accuracy, Visual Grounding Quality); Gemini-3-Flash demonstrated strongest overall performance.

(*Published in ACL Findings 2025—Acknowledged contributor for large-scale data collection.* paper link)

## PUBLICATIONS

Kowshik, M. Ashikuzzaman, Yeasin Arafat Pritom, Md Sohanur Rahman, **Ali Akbar**, and Md Atiqur Rahman Ahad. **"Nurse Care Activity Recognition from Accelerometer Sensor Data Using Fourier-and Wavelet-based Features"** *In Adjunct Proceedings of the 2021 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2021 ACM International Symposium on Wearable Computers, pp. 434-439. 2021.*

## ACHIEVEMENTS & AWARDS

- *Runner up* at Third Nurse Care Activity Recognition Challenge, 2021, awarded by HASCA workshop, ACM Ubicomp. (**Certificate**)
- *Participated* Kharagpur Data Science Hackathon, 2021.

## LEADERSHIP & ACTIVITIES

**President** **Dec 2022 - Feb 2024**
*Luminator, Dr. Muhammad Shahidullah Hall Students' Association.*

**Joint General Secretary** **Apr 2023 - Oct 2024**
*Idrakpur Bus Committee, University of Dhaka.*

**Organizing Secretary** **May 2023 - Aug 2024**
*Students' Association of Munshiganj-Bikrampur, University of Dhaka.*